



CODEX

ZHIIHAO YIN

EMBEDDED & IOT ENGINEER

M.Sc. in HCI & B.Eng. in IoT. Expert in C/C++ embedded programming, smart sensing data acquisition, and multi-protocol edge communications, with excellent hands-on hardware/software co-debugging and industrial validation experiences.

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DEGREE: M.Sc. in HCI
GRAD: Dec 2025

HONORS & SKILLS

Competitions & Certifications

Grand Prize, National Final, 17th Bochuang Cup Embedded Design Competition

1st Prize, National Final, Computer Network Technology Challenge

3rd Prize, National Final, 15th Chinese College Student Computer Design Competition

2nd Prize, Liaoning Computer Design Competition

CET-6 | IELTS 6.5. Fluent in reading technical datasheets, schematics, and English documentations.

Skill Radar

Embedded Systems & HW

C/C++ Programming MCU Drivers ESP32 / STM32
GPIO/UART/SPI I2C/ADC/PWM RTOS Systems Git Control
Oscilloscopes Resource Constraints Datasheet Analysis

Edge-Side Communications & Integration

MQTT Protocol Modbus Protocol TCP/IP & WebSockets
Edge Intelligence HW/SW Co-Debugging
Industrial Field Testing FSM Firmwares
Technical Specifications

Core Attributes & Teamwork

EHS Hazard Inspections Cross-Functional Collab.
Abstraction & Logic Milestone Reviews
Standard Work Instructions English Datasheet Reading

EDUCATION

2024.09 - 2025.09 UNIVERSITY OF NOTTINGHAM (UK)

HCI

M.Sc. in Human-Computer Interaction | Master's Degree

Core Courses: Smart Spatial Interaction, Mixed Reality (MR) Integration, Edge Intelligence Computing, Advanced HW/SW Co-Debugging, Usability Evaluation under Resource Constraints, Human Performance Research.

- Academic Project - VR Rhythm Game Spatial Prototype: Headed development of an audio-reactive VR spatial interaction prototype; implemented C++ multi-threaded data filtering and low-latency serial UART communications to achieve real-time sensor data fusion under strict MCU memory constraints, isolating latency bottlenecks.
- Academic Project - Sensor Data Modeling & Technical Spec Writing: Led capture of high-frequency physical motion sensors. Formulated interaction paradigms, drafted technical specifications, and coordinated milestone reviews to close hardware loop issues.

IOT

2020.09 - 2024.07 DALIAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

B.Eng. in Internet of Things Engineering | Bachelor's Degree

Ranked 1st | GPA 4.1/5.0

Academic Honors: First-Class Golden University Scholarship recipient.

Core Courses: Embedded System Design, MCU Principles & Peripherals (GPIO/UART/SPI/I2C/ADC), RTOS (Real-Time Operating Systems), C/C++ Programming, Wireless Sensor Networks (WSN).

- Academic Project - IoT Industrial Production Line Management System: Independently designed and implemented an industrial production data node on ESP32 microcontrollers. Programmed C/C++ drivers for peripheral interfaces, managed multi-thread scheduling in RTOS, and established low-latency MQTT channels, logging all debugging logs and change histories.

WORK EXPERIENCE

2026.02 - 2026.05 MIAO INC. (SHANGHAI)

EMBEDDED / IOT

Embedded Systems & IoT R&D Engineer

Core Responsibilities: Directed edge co-debugging pipelines, C++ state machine firmwares, and multi-protocol wired/wireless sensing data acquisitions for multiplayer titles "Five-Star Citizen" and "Bombing Blind Shot".

- Edge-Side Data Acquisition & Multi-Protocol Debugging: Collaborated with hardware and QA teams to co-debug sensing prototypes. Created low-level serial (UART/SPI/I2C) and high-level TCP/UDP wireless transmitters to aggregate physical inputs, filing standard change logs.
- "Bombing Blind Shot" HW/SW Co-Debugging & Edge Intelligence: Maintained edge node processing pipelines and custom MCP server communication middlewares connecting local LLM agents to Roblox runtimes. Directed integration tests and pilot validation, securing high reliability.
- Git Version Management & Technical Review Standards: Supervised branch deployments and code reviews using Git. Programmed automated regression scripts to guarantee microcontroller firmware compatibility, ensuring fast resolution and issue closures.

SERVER DEV

2023.09 - 2024.02

SHANGHAI HUICHENG TECHNOLOGY CO., LTD. (DIGITAL TWIN R&D CENTER)

Embedded & Server R&D Intern

- Edge Gateway Design & Modbus/MQTT Integration: Assisted in configuring edge gateway protocols (Modbus, MQTT) and peripheral MCU drivers. Analyzed high-throughput industrial telemetry logs, supporting field pilot testing and deployment validation.
- Code Quality Standards & Version Controls: Strictly adhered to coding specifications and branch merge protocols; optimized telemetry buffer queues to resolve concurrency lags, authoring 10+ interface specs and code reviews.

PROJECT EXPERIENCE

2025.01 - 2025.06 WARHAMMER DIGITAL LINE USER RESEARCH

USER RESEARCH

System Usability & Co-Debugging Analyst (University of Nottingham Campus Collaboration Project)

Responsibilities: Led player research for the Games Workshop Warhammer digital ecosystem. Utilized mixed-method studies to diagnose and optimize interactive UI flow pain points.

- Multi-Platform Communication & Interface Auditing: Supervised API network lag and protocol throughput audits between mobile interfaces and backend servers, locating menu nesting and transmission bottlenecks to draft optimization documents.
- Cross-Functional Review & UGUI Optimization: Led design and milestone reviews with the Games Workshop digital division, pushing interface redesigns and progressive onboarding flows to lower mobile 3-day player churn by 25%.

UNITY / C#

2025.03 - 2025.06 "MILLENNIUM MIST" - GAMEPLAY VALIDATION PROTOTYPE

Solo Unity & C++ Engine Prototype Developer

Responsibilities: Built an action-shmup prototype from scratch in C++/C# to validate high-performance business logics and combat frameworks under resource constraints.

- C++ Business Logic & HFSM Firmware Design: Programmed Hierarchical Finite State Machines (HFSM) to guide complex combat and AI rules, maximizing memory allocation and CPU efficiency under constrained resources.
- Tactile Actuator Calibration & Performance Benchmarking: Synchronized external tactile feedback, HUD layouts, and VFX triggers; optimized garbage collection and buffer pools to handle high-frequency particle surges, maintaining stable frame rates under peak stress.

2025.07 - 2025.10

"MR RECIPES" - MIXED REALITY SPATIAL INTERACTION PROTOTYPE

XR / HCI

Meta Quest 3 Smart Sensing & Spatial Positioning R&D

Responsibilities:

Built a spatial cooking assistant for Meta Quest 3 combining sensor inputs and real-time physical overlays.

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Sensor Calibration & Spatial Anchor Alignments:

Deployed eye-gaze and hand-tracking spatial sensors in Meta Quest 3. Debugged virtual-physical boundaries in real kitchen setups, resolving background noise to anchor UI elements at sub-millisecond rates.

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Pilot Onsite Testing & Defect Loop Closures:

Conducted contextual inquiries and pilot runs in target environments; solved hardware failures (due to steam and wet hands) by writing robust fault-tolerance algorithms in solid firmware scripts.

ACTIVITIES

STAGE SUPPORT

Long-term Active

LOGISTICS & STAGE SUPPORT VOLUNTEER

Prominent Music Festivals (Midi, Aranya, Bubble Land, etc.)

Responsibilities:

Worked backstage under rigorous EHS guidelines to enforce safety standards, evaluate electricity risks, and maintain zero-accident operations.

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Safety Audits & Lightning Proofing Suggestions:

Actively inspected backstage power cabinets and wiring layouts; proposed 10+ lightning and electrical leak prevention protocols to ensure absolute event security.

2022.06 - 2023.04

PRESIDENT OF MAKER SPACE

MAKER SPACE

University Interdisciplinary Technology Association

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Embedded Lab Instruction & Lab Standardizations:

Supervised hardware selections, I/O pin layouts, and serial UART communication firmwares on ESP32/STM32 platforms for 100+ members. Secured sponsorships to enforce laboratory safety guidelines.